

Inayat Ali

PhD, SMIEEE

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Career Objective

Holding a Ph.D. degree from the [Korea University of Science and Technology \(UST\)-KISTI Campus](#) and possessing over ten years of research experience in computer networks, I aim to contribute as a researcher in **programmable networks**, **future internet**, **internet protocol**, and **resource allocation for application performance**.

UST is a Federated Research University where PhD candidates are selected to work at Korea's Premier National Laboratories.

Employment History

Electronics and Telecommunications Research Institute (ETRI)

Daejeon, S.Korea

POST-DOCTORAL RESEARCHER

May. 2022 ~Present

- ETRI is a premier national research institute in Korea, where I design network architectures and transport protocols, focusing on AI-driven optimization, programmable networks, and cloud/edge resource allocation.

Korea Institute of Science and Technology Information (KISTI)

Daejeon, S.Korea

GRADUATE STUDENT RESEARCHER

Sept. 2017 ~Feb. 2022

- Worked as a full-time student researcher at KISTI, a premier center for supercomputing and data science, contributing to multiple computer networking projects and publishing in international and national journals and conferences.

Comwave Institute of Sciences and Information Technology

Abbottabad, Pakistan

VISITING LECTURER

Sept. 2015 ~Jan. 2017

- I worked as a visiting lecturer, teaching undergraduate computer science students. The courses I taught included Introduction to Telecommunication Systems, Computer Networks and Communications, and Object-Oriented Programming.

COMSATS University of Science and Technology, Abbottabad, Pakistan

Abbottabad, Pakistan

RESEARCH ASSOCIATE

Sept. 2015 ~Jan. 2017

- I worked as a Research Associate, teaching undergraduate computer science students and conducting research in programmable networks.

Education

Ph.D in Data & HPC Science

Korea University of Science and Technology (UST) 

Korea Institute of Science and Technology Information. KISTI School 

Daejeon, South Korea

MASTER PHD COMBINED PROGRAM

Sept. 2017 ~Feb. 2022

- CGPA. 4.15/4.5
- Thesis Title: Efficient Content Retrieval in Named Data Networking Based Highly Dynamic Wireless Environment

B.S. Telecommunication and Networking

COMSATS University of Science and Technology, Islamabad, Pakistan. 

Abbottabad, Pakistan

FOUR YEARS HONOR DEGREE

Feb. 2011 ~March. 2015

- Distinction: Academic Silver medal
- CGPA. 3.61/4.0
- Thesis title: Policy Based Routing Using Software Defined Networking (SDN).

RESEARCH INTEREST

- Programmable Networks, Cloud and HPC, Future Internet, Information-Centric Networks, Internet Protocol, and Resource allocation for improving immersive Internet application performance.

Research Publications

- For full citations, please visit my [Google Scholar](#).
- I. Ali, S. Hong, P.-K. Park, and R. Ryu, "Stability-Aware Learning-Based Active Queue Management in AI Hyperscale Data Centers". *"IEEE LCN 2026"* Oct 7-9, Coimbra, Portugal 2026. **Accepted**

3. S. Hong, H. Yoon, S. Moon, **I. Ali**, “Latency in Real-Time 3D Volumetric Streaming: A Comprehensive Study”. “*The 17th International Conference on Ubiquitous and Future Networks*” July 7 - 10, Milan, Italy 2026. <https://arxiv.org/pdf/2605.22131>
4. U. Islam, D. Din, A. Huq, **I. Ali**. “Federated Security Framework: A Heterogeneous Federated Learning Architecture for Privacy-Preserving Intrusion Detection in IoT Networks”. *Transactions on Emerging Telecommunications Technologies*, vol. 37. Issue. 4. 2026. doi. 10.1002/ett.70402
5. M. Iqbal, S. Waheed, **I. Ali**, S Z. Shah, S A S. Bokhari, S. Sultan. “Event-driven localization in wireless sensor networks: a machine learning approach”. *The Journal of Supercomputing (Springer Nature)*, vol. 82. Issue. 3. 2026. doi. 10.1007/s11227-026-08273-3
6. **I. Ali**, S. Hong, and T. Kim, “A Multilevel Network-Assisted Congestion Feedback Mechanism for Network Congestion Control”. *Computers and Electrical Engineering (Elsevier)*, vol. 123. pages. 110067. 2025. doi. 10.1016/j.compeleceng.2025.110067
7. **I. Ali**, S. Hong, and T. Cheung, “Quality of Service and Congestion Control in Software-Defined Networking Using Policy-Based Routing”. “*Applied Sciences*”, vol 14, issue 19, 2024. doi. 10.3390/app14199066
8. **I. Ali**, S. Hong, P.-K. Park, and T. Kim, “Rethinking Explicit Congestion Notification: A Multilevel Congestion Feedback Perspective”. “*ACM NOSSDAV 2024*” April 15 - 18, Bari, Italy 2024. doi. 10.1145/3651863.3651885
9. **I. Ali**, S. Hong, and T. Cheung, “Congestion or No Congestion: Packet Loss Identification and Prediction Using Machine Learning”. “*IEEE PlatCon 2024*” August 26 - 28, Jeju Island, Korea 2024. doi. 10.48550/arXiv.2408.03007
10. **I. Ali**, S. Hong, P.-k. Park, and T. Y. Kim, “Performance Evaluation of Transport Protocols and Roadmap to a High-Performance Transport Design for Immersive Applications”, *International Conference on Ubiquitous and Future Networks (ICUFN)*, July 4 ~7 Paris, France, 2023. arXiv preprint arXiv:2306.16692 (2023). doi. 10.1109/ICUFN57995.2023.10199443
11. **I. Ali** and H. Lim, “NameCent: Name Centrality Based Data Broadcast Mitigation in Vehicular Named Data Networks,” in *IEEE Access*, vol. 9, pp. 162438 ~162447, 2021. doi. 10.1109/ACCESS.2021.3133016
12. **I. Ali** and H. Lim, “Anchor-Less Producer Mobility Management in Named Data Networking for Real-Time Multimedia”, *Mobile Information System*, vol. 2019, Article ID 3531567, 12 pages, 2019. doi. 10.1155/2019/3531567
13. **I. Ali** and H. Lim, “Controlled Data Broadcast Storm in Vehicular Named Data Networking”, *13th International Conference on Internet (ICONI 2021)*, Dec 12 ~14, Jeju, South Korea.
14. **I. Ali** and H. Lim, “Cache Performance Investigation in NDN for Large Scale Scientific Data”, *The Journal of Korean Institute of Communications and Information Sciences*, vol 44. no. 10. pp. 1860 ~1865, 2019. doi. 10.7840/kics.2019.44.10.1860.
15. **I. Ali**, S. Sabir, and Z. Ullah, “Internet of Things Security, Device Authentication and Access Control: A Review,” *International Journal of Computer Science and Information Security*, vol. 14, no. 8, pp. 456 ~466, 2016. doi. 10.48550/arXiv.1901.07309
16. **I. Ali**, E. Khan, and S. Sabir, “Privacy-preserving data aggregation in resource-constrained sensor nodes in the Internet of Things: A review”, *Future Computing and Informatics Journal*, vol. 3, no. 1, pp. 41 ~50, 2018. doi. 10.1016/j.fcij.2017.11.004
17. S. Sabir, **I. Ali**, and E. Khan, “H-LPS: a hybrid approach for user’s location privacy in location-based services,” *International Journal of Ad Hoc and Ubiquitous Computing*, vol. 32, no. 3, pp. 171 v 179, 2019. doi. 10.1504/IJAHUC.2019.103142

Patents

18. L. Huhnkuk and **I. Ali**, “Forwarding method and system for data broadcast reduction based on name centrality and applied in vehicle named data networking”. *US Patent no. 20230217522 A1*. Publication date. 2023/7/6. Application granted: Nov, 2025. Link <https://patents.google.com/patent/US12464588B2/en>
19. L. Huhnkuk and **I. Ali**, “NAME-CENTRICITY-BASED BROADCAST MITIGATION FORWARDING METHOD AND SYSTEM APPLIED IN VNDN”. *KR Patent no. 1026688250000*, Application no. 1020210192759, Registration no. 1026688250000, Reg Date. 2024.05.20. Doi. <https://doi.org/10.8080/1020210192759>
20. P. Park, B. Kwak, M. Seong, **I. Ali**, Y. H Sun, S.W Hong. “APPARATUS AND METHOD FOR CONGESTION CONTROL MECHANISM BASED ON REINFORCEMENT LEARNING TO SUPPORT ULTRA-LOW-LATENCY INTERACTIVE SERVICES IN COMMUNICATION SYSTEM”. *Application no. 1020240001562*. Application Date. 2024.01.04. Publication no. 1020250106938. Publication Date. 2025.07.11. Doi. <https://doi.org/10.8080/1020240001562>

Skills

Programming	C++, Python, C-sharp, Network Programming, P4
Networking Tools	NS-3, Mininet, ndnSIM, Packet Tracer, Linux
Languages	English (Proficient, IELTS (C1) bands 7), Urdu (Native), Korean (beginner)

Projects

Resource Orchestration and Auto-scaling for Guaranteed Application Performance

ETRI, South Korea

GOALS

April. 2025 ~PRESENT

- Research and develop an intelligent orchestrator for guaranteed application performance
- Develop an auto-scaling system to ensure application requirements

Design of High Performance, Low Latency Internet Transport Protocol

ETRI, South Korea

GOALS

- Research and develop a new transport layer protocol for challenged networks.
- Design, simulate, and verify the proof of concept.

May. 2022 ~PRESENT

Efficient Content Retrieval in Named Data Networking Based Vehicular ad-hoc Network

KISTI, South Korea

GOALS

- Explore NDN for Internet of Vehicles (IoV).
- Design and implement solution for mitigating data broadcast storm problem in NDN-based IoV.

Jan. 2020 ~Feb 2022

Mobility Management in Named Data Networking

KISTI, South Korea

GOALS

- Study mobility support in NDN and ICN architectures
- Design and implement solution for mobility support in NDN-based networks.

Sep. 2017 ~Dec 2019

Policy Based Routing Using Software Defined Networking (SDN)

COMSATS, Abbottabad

GOALS

- Explore Policy-based routing in SDN-based campus network.
- Design policy to prioritize RTP traffic during an online event on the campus.

Feb. 2014 ~Jan 2015

Honors, Awards & MOOCs

HONORS & AWARDS

- 2015 **Academic Silver Medal**, B.S. Graduation
- 2014 **Second prize**, Semester Project Competition
- 2016 **Scholarship**, Master studies
- 2017 **Ph.D. Scholarship**, UST Korea

Abbottabad, Pakistan

Abbottabad, Pakistan

COMSATS, Pakistan

Daejeon, S. Korea

MOOCs AND CERTIFICATES

- Machine Learning by Andrew Ng (Coursera)
- Network Protocol Development in C (from scratch) (Udemy)
- Linux Socket Programming Hands On - Zero to Hero (Udemy)

Peer Review Activity

Reviewer for journals

I served as a reviewer for the following journals: For full review history, visit my ORCID profile.

<https://orcid.org/0000-0001-8847-294X>

IEEE Internet of Things Journal, IEEE Access, Computer Communications, IEEE Transactions on Network Science and Engineering, Journal of Supercomputing and Scientific Reports.

References

- Available on Request